

SOLUTION

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI
 Second Semester, Session: 2025 – 2026
 Control Systems (ECE/EEE/INSTR F242)

MM: 06

Time: 12 Minutes

Quiz - 5

Date 07.04.2026

Note: Write your final answers in the space provided only. Show all the necessary steps in solution.

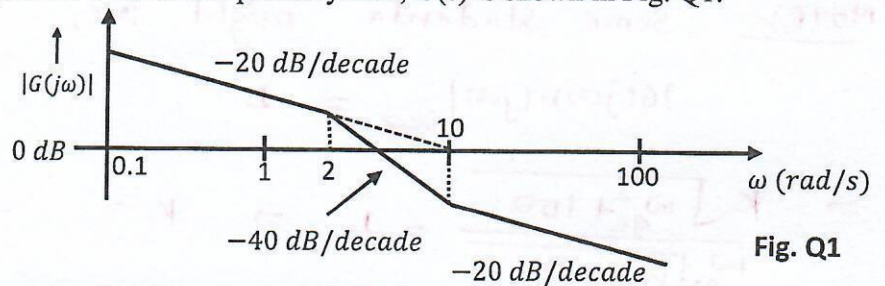
ID: **Solution**

Name: **Solution**

Section:

1. The asymptotic Bode magnitude plot for a minimum phase system, $G(s)$ is shown in Fig. Q1.

$$G(s) = \frac{K(s+a)}{s(s+b)}$$



(A) Draw the pole-zero plot of this system in the space provided (Fig. A1).

(B) Find the value of 'K' for this system $G(s)$.

(C) If the gain crossover frequency of this system is 4.5 rad/s, find the phase margin in degrees. Assume unity feedback system.

As observed from bode magnitude plot,

$$|G(j\omega)|_{\omega=0.1} = 40 \text{ dB}$$

1. Initially the slopes are changing (-20 dB/decade to -40 dB/decade) at $\omega = 2 \text{ rad/s}$ so it should correspond to a pole.

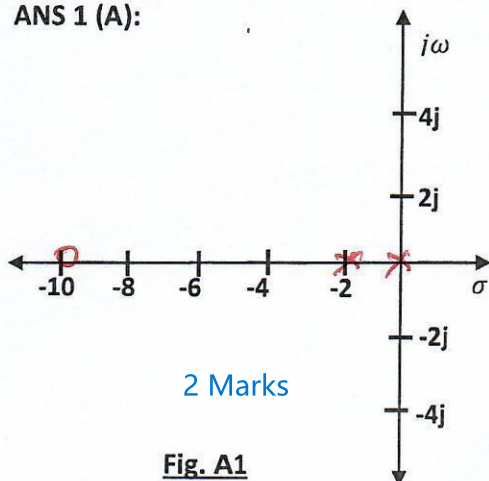
2. at $\omega = 10 \text{ rad/s}$, slope increases by +20 dB/decade so it should correspond to a zero.

$$\text{So, } G(s) = \frac{K'(1+s/\omega_z)}{s(1+s/\omega_p)} = \frac{K'(1+s/10)}{s(1+s/2)}$$

$$\text{Also, } -20 \log(K') - 20 \log(0.1) = 40 \text{ dB} \\ \Rightarrow K' = 10$$

$$\text{So, } G(s) = \frac{10(1+s/10)}{s(1+s/2)} = \frac{2(1+s/10)}{s(1+s/2)} \\ \Rightarrow \boxed{K=2}$$

ANS 1 (A):



2 Marks

Fig. A1

ANS 1(B): $K = 2$ 2 Marks

ANS 1(C): Phase Margin (PM) = 48.2° 2 Marks

$$P.M. = 180^\circ + \angle G(j\omega)H(j\omega)|_{\omega_{gc}}$$

$$\angle G(j\omega)H(j\omega) = -90^\circ + \tan^{-1}(\omega/10) - \tan^{-1}(\frac{\omega}{2})$$

$$= -131.80^\circ$$

$$\text{So, } \boxed{P.M. = 48.2^\circ}$$

--- not

* NOTE:- Some students might use,

$$|G(j\omega)H(j\omega)|_{\omega_{gc}} = 1$$

$$\Rightarrow \frac{K \sqrt{\omega_{gc}^2 + 100}}{\omega_{gc} \sqrt{\omega_{gc}^2 + 4}} = 1 \Rightarrow K = \frac{\omega_{gc} \sqrt{4 + \omega_{gc}^2}}{\sqrt{100 + \omega_{gc}^2}}$$

$$\boxed{K = 2.02}$$

Please award marks for this approach also, ---